

Natural Language Processing

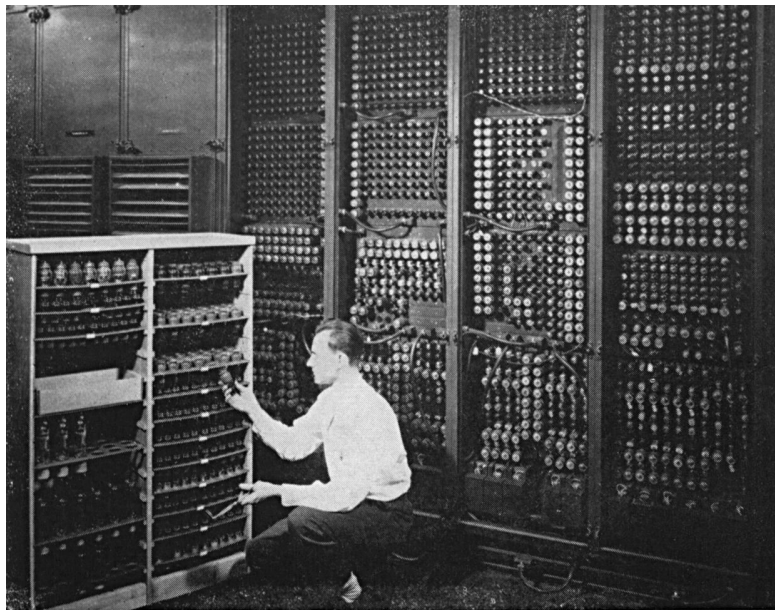
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Introduction and History

Natural Language Processing

- Multidisciplinary field related to linguistics, computer science, artificial intelligence, and human-computer interaction (HCI).
- It emerged around 1950
- Breakthrough in NLP in the recent years due to
 - fast computer hardware like GPUs (graphics processing units),
 - Techniques like deep learning.

https://en.wikipedia.org/wiki/Computer#/media/File:ENIAC-changing_a_tube.jpg



Replacing a bad tube meant checking among ENIAC's 19,000 possibilities.

ELIZA (1964-1966)

<https://en.wikipedia.org/wiki/ELIZA>

- ELIZA was an early chatbot system that convinced users that they were talking to an actual human
- DOCTOR, simulated a psychotherapist by simply parroting back at patients what they had just said

Welcome to

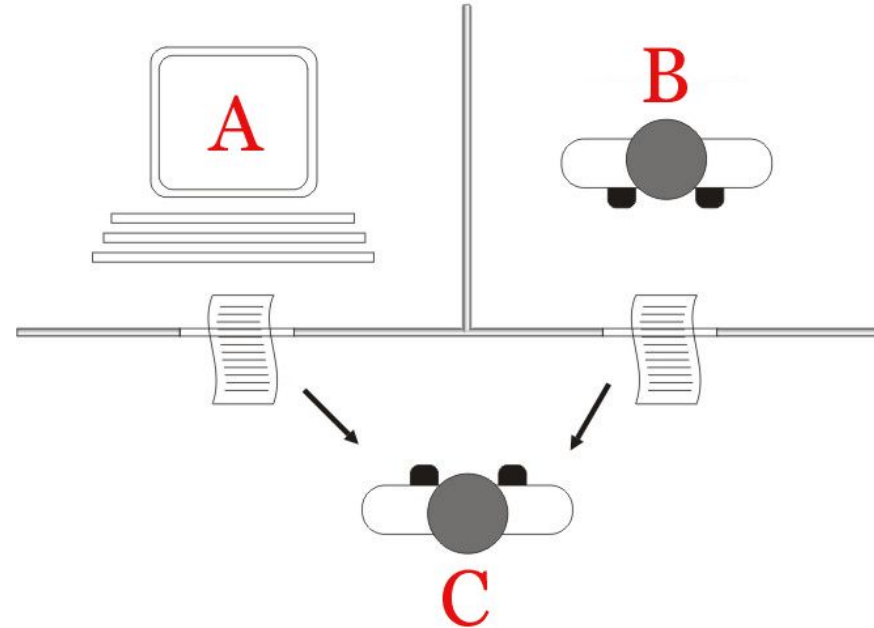
EEEEEE	LL	IIII	ZZZZZZ	AAAAA
EE	LL	II	ZZ	AA AA
EEEE	LL	II	ZZZ	AAAAAAA
EE	LL	II	ZZ	AA AA
EEEEEE	LLLLLL	IIII	ZZZZZZ	AA AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU: Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU: They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU: Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU: He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU: It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:

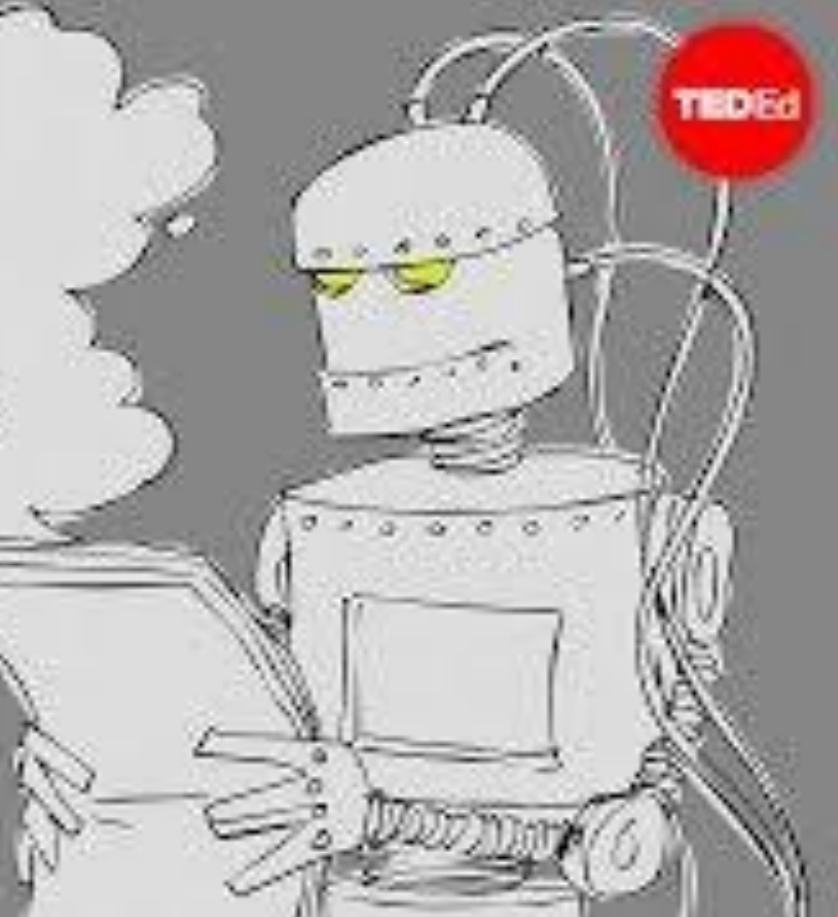
Turing Test

- Criterion of intelligence
 - Can a machine exhibit intelligent behavior indistinguishable from that of a human
- Is it possible to distinguish between human-to-human and human-to-machine natural language conversations?



CAN A
COMPUTER
TALK LIKE A
HUMAN?

TEDEd



NLP

- Processing and understanding natural language content such as speech and text.
- Speech recognition, natural language understanding (NLU), or natural language generation (NLG)

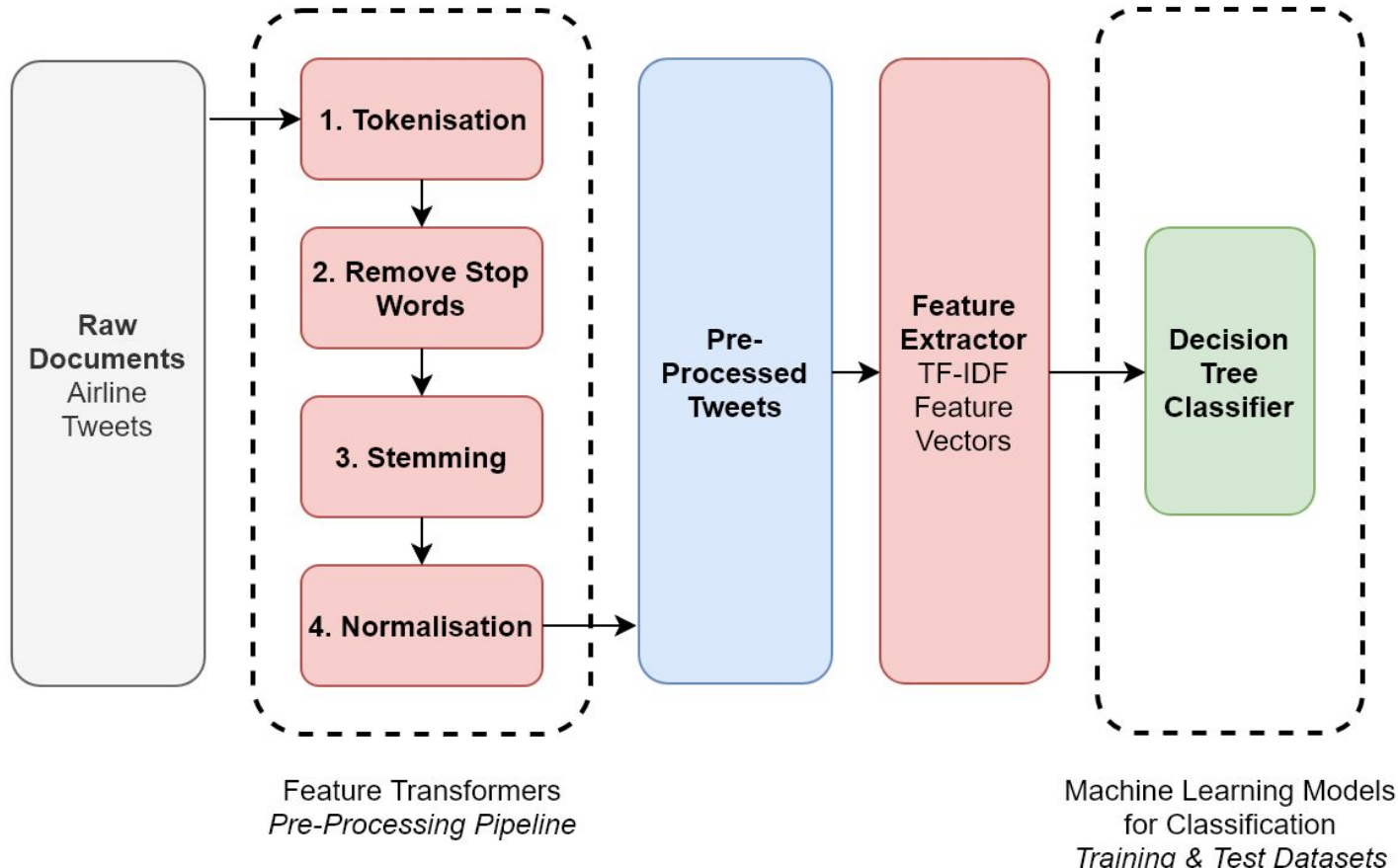
Applications

- Sentiment analysis:
- Text summarization
- Book generation:
 - In 2018, the novel 1 the Road was published as the first work by a neural network (Merchant, B., 2018).
- Machine translation
- Answering questions
- Fake news detection
- Chatterbots
- Clinical text mining
 - applying NLP on patient history documents and assessments to extract insights

NLP Pipeline

NLP Pipeline

<https://www.oreilly.com/library/view/machine-learning-with/9781789346565/def879fb-6164-4f84-992b-d61f5f7d5a83.xhtml>



Word Tokenization

- Break this sentence down into units like words

The leaves of these trees are green during the spring and summer.

['The', 'leaves', 'of', 'these', 'trees', 'are', 'green', 'during', 'the', 'spring', 'and', 'summer']

Stop Word Removal

- A stop word is a commonly used word such as “the”, “a”, “is”, “are” that we can ignore and remove because these words are not useful for our task.
- There is no universal or common list of “stop words” used by all NLP tools.
- Stop words might differ from one application to another

[‘leaves’, ‘trees’, ‘green’, ‘during’, ‘spring’, ‘summer’]

Stemming and Lemmatization

- Replace each word with its root
- Converting words to their canonical form
- Stemming: removing the suffix (the result might not be a dictionary word):

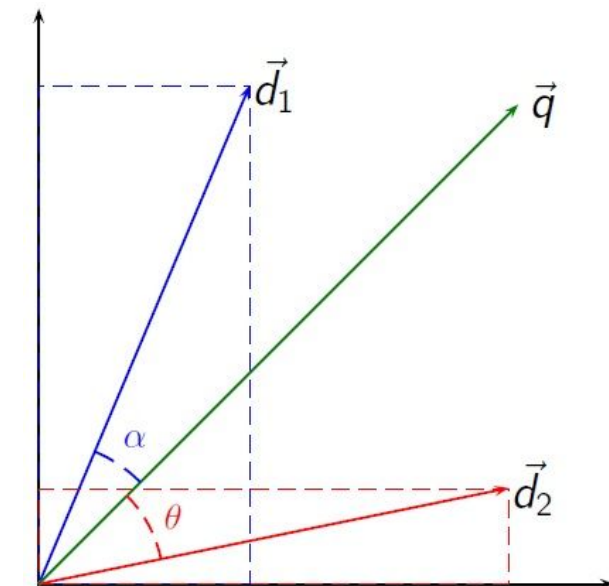
['leav', 'tree', 'green', 'during', 'spring', 'summer']

- leav is not a valid word and doesn't exist in dictionaries. we can apply lemmatization if we need valid root words:

['leaf', 'tree', 'green', 'during', 'spring', 'summer']

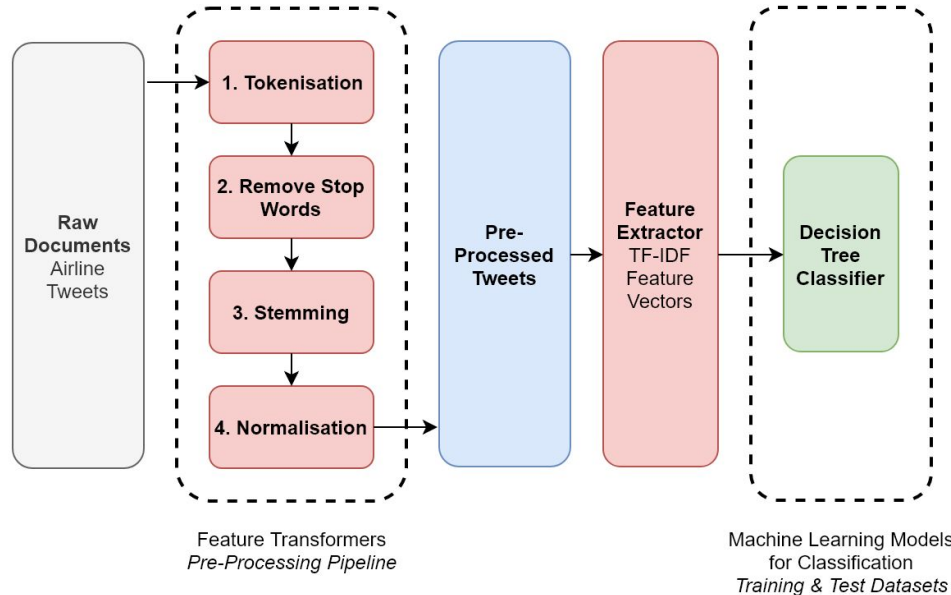
Vector Spaces

- Computers can only understand numbers and digits
- We can't feed words into machine learning algorithms directly and we have to need to convert words into numbers by a process called vectorization.



Training the Model

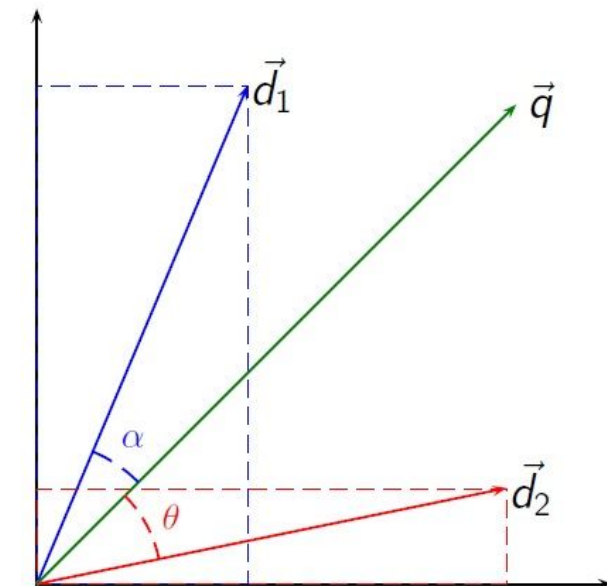
- At this step, we train an model such as decision tree for a task in hand
- The vectors will be the input to this model



Vectorization

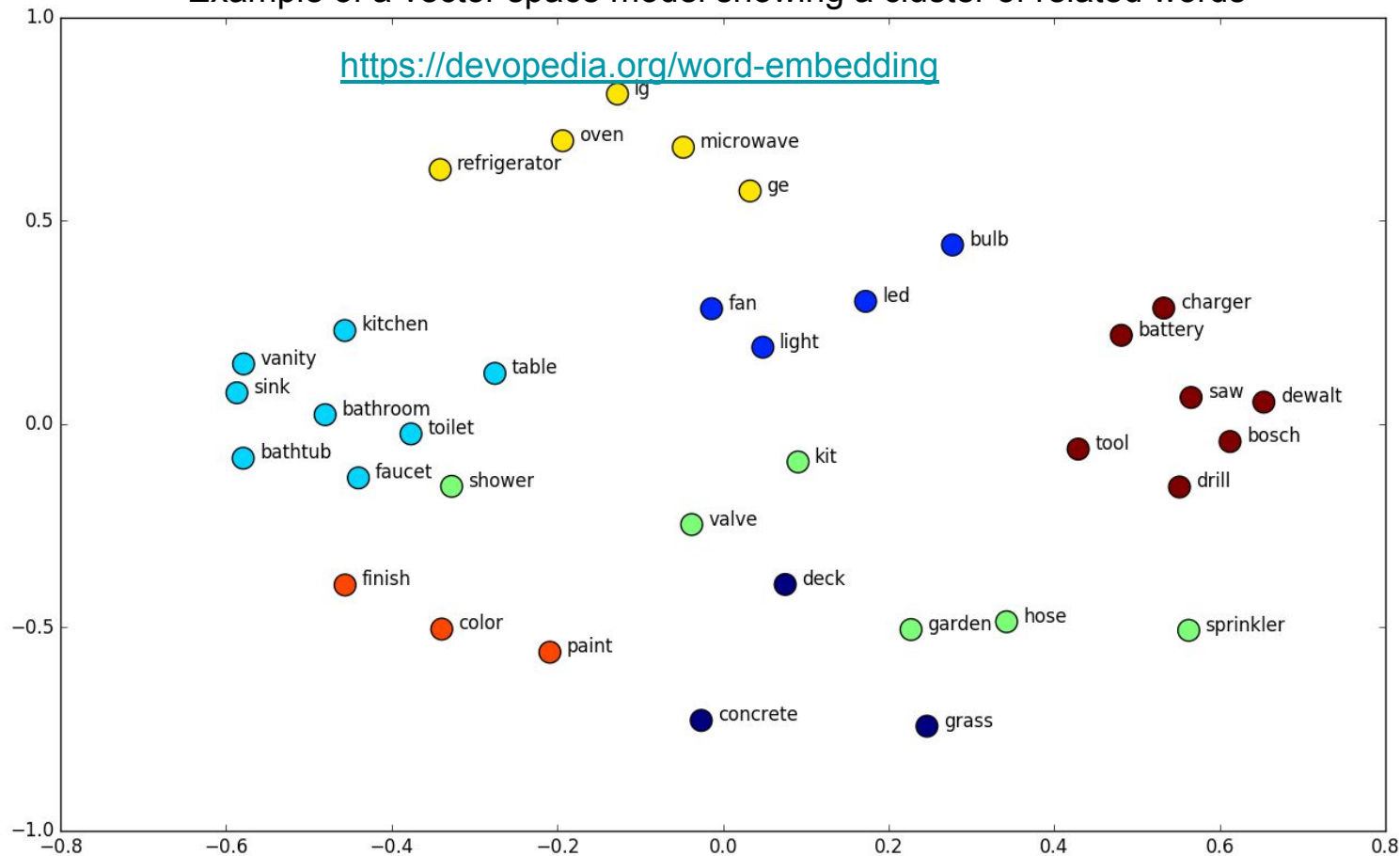
Vector spaces

- The cosine of the angle two vectors=similarity between the two vector.
- Figure below shows a text search query from a collection of documents.
Relevant documents vectors have a smaller angle with the query vector



$$\cos(d_j, q) = \frac{d_j \cdot q}{\|d_j\| \|q\|} = \frac{\sum_{i=1}^N w_{ij} w_{iq}}{\sqrt{\sum_{i=1}^N w_{ij}^2} \sqrt{\sum_{i=1}^N w_{iq}^2}}$$

Example of a vector space model showing a cluster of related words



One-Hot Encoding

- The most basic text vectorization technique
- Each word is one dimension in the vector
- If that word appears in the text, the corresponding value =1 else =0

Corresponding word	One-hot encoding vector
spring	1
summer	1
during	1
green	1
like	0
my	0
tree	1
cat	0
dog	0
red	0
leaf	1
...	0

Bag-of-Words

- Similar to one-hot encoding (each row in the vector corresponds to one word)
- In contrast to One-hot encoding, it also considers the number of times a word appears
- **Problem:**
 - high count for a word != word importance.
 - For instance, common words like “want” and “is” have high frequency, but that doesn't indicate importance.

Corresponding word	One-hot encoding vector
spring	2
summer	10
during	3
green	1
like	4
my	0
tree	2
cat	0
dog	3
red	0
leaf	1
...	0

TF-IDF (Term Frequency, Inverse Document Frequency)

- Normalize the term frequencies by multiplying it by the inverse of document frequency (This addresses the count=importance problem in BoW model)
- The tf-idf value is offset by the number of documents in the entire corpus that contain the word.

$$\text{tfidf}(t, d, D) = \text{tf}(t, d) \cdot \text{idf}(t, D)$$

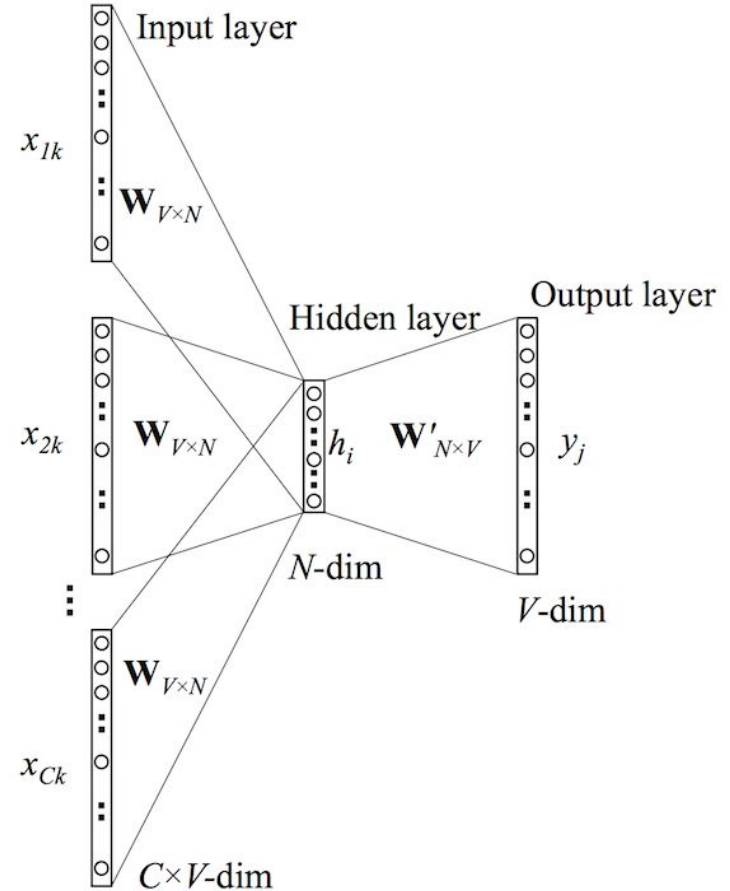
$$\text{IDF}(t, D) = \log(N/n_t)$$

Word Embeddings

- One-hot encoding, BoW, TF-IDF disregards the grammar or ordering of the words
- Word embeddings are real-valued vectors that encode the meaning of words
- Words that have similar meaning are closer in the vector space
- Lower dimensional and dense space
 - In previous methods the size of vector was equal to the size of dictionary
- Since it converts the original space with many dimensions per word to a lower dimensional, dense, and continuous vector space.
- Word embeddings consider the context of a word (i.e., other words that accompany a word)

Word2Vec

- Uses shallow neural networks with one hidden layer and one output layer.
- For example, one neural network architecture to generate Word2vec embeddings is called CBOW (continuous bag-of-words) as shown here.

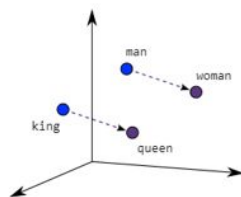


- Input: the one-hot encoding of words surrounding a given word ($x_{1k}, x_{2k}, \dots, x_{Ck}$).
 - Each surrounding word is then converted to a V -dimensional one-hot encoding vector (assuming there are V words in the dictionary)
 - Since we have C words, our input vector will be $C \times V$ elements.
- The expected output vector (y_j) is a V -dimensional one-hot encoding for the target word.
- There is one hidden layer with N units
- Each input word vector is connected to this hidden layer with a weight matrix of W of $V \times N$ dimensions.
- the C N -dimensional vectors from the hidden layer will be averaged element-wise and the result will be the activation layer to be fed into the final softmax layer.
- After the training is done, the weights from the hidden layer can be used to generate a Word2vec vector of size N for any given word.

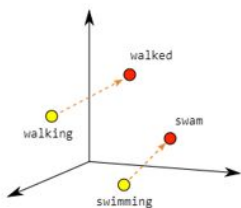
Word2Vec

<https://towardsdatascience.com/word-embeddings-for-nlp-5b72991e01d4>

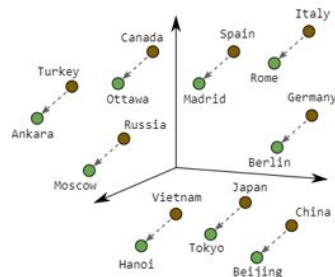
Word2Vec



Male-Female

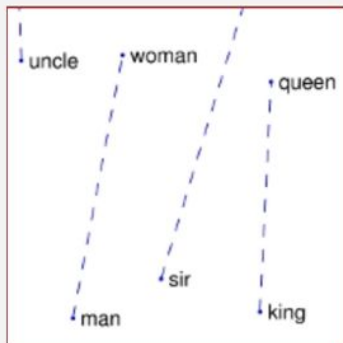


Verb Tense

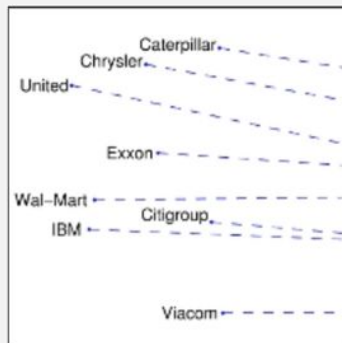


Country-Capital

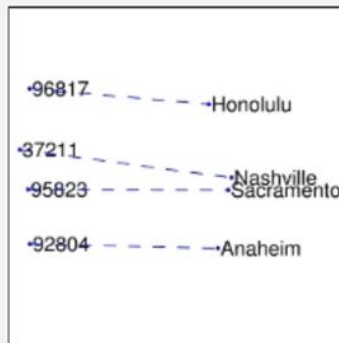
GloVe



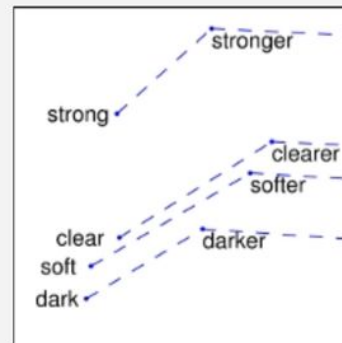
man - woman



company - ceo



city - zip code



comparative - superlative

Sentiment Analysis

Sentiment Analysis

- Also called opinion mining
- Without an automated sentiment analysis tool, it would be impossible to analyze billions of comments and reviews written by online users every day.
- To enhance customer experience by measuring the overall sentiment of users towards their products.
- Classify the polarity of a given text as positive, negative, or neutral, or return a score indicating how positive or negative the text is (for example, on a scale from -10 to +10)

Subjective Text

- Sentiment analysis is mainly intended for text which is subjective (rather than objective)
 - A 'subjective' text contains personal opinions or judgments versus factual information.
 - Subjective text is usually expressed with emotions or feelings.

“Summers in California are hot.”

“I hate this hot weather, I prefer the winter cold!”

Supervised Sentiment Analysis Model

- Input: textual data with a polarity label
- Build a machine learning model that can classify the sentiment of future input texts to the polarity labels
- Similar to a typical machine learning model

Named Entity Recognition

Named Entity Recognition

- Locate and classify named entities in a text and classify them into categories

“California State University, Long Beach was established in 1949 as a public research university in Long Beach, California. CSULB is one of the largest universities in the state of California by enrollment having 39,435 enrolled students for the fall 2021 semester.”

- California State University, Long Beach: organization name
- Long Beach: geopolitical entity
- California: geopolitical entity
- 1949: date

- Introduction to Word Embeddings
- Vector Spaces and Word embeddings
- Word Embeddings: IMDB Sentiment analysis
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